METHODOLOGY WHITE PAPER

GEMS 2026

Gaming Economic Modeling System

The Casino Economic Impact Model

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Abstract

GEMS (Gaming Economic Modeling System) is a state-level economic impact model for casino and gaming operations covering all 50 US states and the District of Columbia. The model applies Input-Output (IO) analysis under the Industry Technology Assumption, using EPA State Input-Output tables, BLS Quarterly Census of Employment and Wages data, and gambling-specific coefficients derived from BEA Detail IO tables. For a given revenue profile, the model estimates direct, indirect, and induced effects on economic output, GDP (value added), employment, wages, and tax revenue. This white paper documents the modeling framework, data sources, and key methodological decisions of the 2026 edition, and describes the model's appropriate uses and limitations.

Suggested citation

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When citing results produced by the model, please also state the analysis date and the inputs used (state, property type, and revenue assumptions), as results depend on user-supplied inputs.

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1. Introduction

Casino developments are routinely evaluated on their wider economic contribution: the jobs they support, the value they add to regional GDP, and the tax revenue they generate. Yet general-purpose economic multipliers are poorly suited to gaming. Published entertainment and recreation multipliers blend casinos with golf courses, fitness centers, and amusement parks — businesses with fundamentally different cost structures and labor intensities.

GEMS addresses this gap. It combines state-specific Input-Output multipliers with gambling-specific industry coefficients and property-type-specific revenue structures to produce defensible, state-level estimates of the economic footprint of casino and gaming operations. The model supports land-based property types (casino hotels, stand-alone casinos, slot parlors, and bar/restaurant gaming) as well as online casino and sports betting operations.

For each analysis, the model reports five metrics — gross output, GDP (value added), employment, wages, and tax revenue — each decomposed into direct, indirect, and induced components with the implied multipliers.

2. Modeling Framework

2.1 Industry Technology Assumption

The model is built on the Industry Technology Assumption (ITA) framework of Input-Output economics. The direct requirements matrix is constructed from the Make and Use tables:

$$\mathbf{A} = \mathbf{D} \times \mathbf{B}$$

where **D** is the market share matrix derived from the Make table (the share of each commodity produced by each industry) and **B** is the commodity coefficient matrix derived from the Use table (the amount of each commodity required per dollar of output). The result **A** is an industry-by-industry direct requirements matrix.

2.2 Leontief Inverse and Type I Multipliers

$$\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$$

The Leontief inverse captures all rounds of inter-industry purchasing. Each column sum gives the Type I output multiplier — total output generated per dollar of direct output, including all supplier (indirect) effects.

2.3 Type II Augmentation

The A matrix is augmented with a household row (labor income coefficients) and household column (consumer spending patterns). The augmented Leontief inverse yields Type II multipliers, which additionally capture induced effects from the spending of wages earned in direct and indirect activity.

3. Effect Decomposition

Effect	Source	Example
Direct	The initial economic activity	Casino hires employees, pays wages, generates revenue
Indirect	Supply-chain purchasing	Casino buys food, linens, and equipment from suppliers; suppliers buy from their suppliers
Induced	Household spending	Casino and supplier employees spend wages on housing, groceries, and services

Effects are decomposed from the multiplier pairs as follows:

```
Direct output = revenue
Indirect output = revenue × (Type I − 1)
Induced output = revenue × (Type II − Type I)
Total output = revenue × Type II
```

The same decomposition applies to GDP, wages, and tax metrics using their respective multiplier pairs.

4. Employment Estimation

Employment is estimated with employment-weighted coefficients rather than output-based multipliers, because a dollar of indirect or induced GDP generates more or fewer jobs depending on whether it flows to labor-intensive sectors (restaurants) or capital-intensive sectors (utilities):

```
Jobs = GDP_effect × Employment_Coefficient
```

Coefficient	Applies to	Reflects
Emp_Coef	Direct employment	Labor intensity of the gaming industry itself
Indirect_Emp_Coef	Indirect employment	Weighted average labor intensity of supplier industries
Induced_Emp_Coef	Induced employment	Weighted average labor intensity of household-spending industries

Coefficients are expressed as jobs per $1M of value added (GDP), not gross output. Using gross output as the denominator would double-count intermediate purchases and understate the jobs-per-dollar ratio. Coefficients are computed from the Leontief inverse weighted by sector-level employment-to-GDP ratios from QCEW data. When users provide actual employment or wage figures, the model uses those values for the direct effect and estimates only the indirect and induced components.

5. Reporting Basis & Inflation Adjustment

All monetary results the model reports — output, GDP, wages, and tax revenue — are in current-year (nominal) dollars, the same dollars the user enters. The dollar multipliers and coefficients are unitless ratios applied directly to current-dollar revenue, so their results are automatically in current dollars. No reported dollar figure is deflated.

Employment is the one quantity that needs a base-year adjustment, because the employment coefficients carry a dollar denomination: they are jobs per $1M of value added expressed in **2019 dollars** (the IO table base year, and the denominator used when the coefficients were estimated). Applying that coefficient to a current-dollar GDP figure would mismatch the price bases. The model therefore restates GDP in 2019 dollars purely as an internal step in the jobs calculation, then applies the coefficient:

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Deflator = CPI_2019 / CPI_current_year
Jobs = (GDP_current × Deflator) × Employment_Coefficient
```

This conversion affects the employment estimate only; it does not change any dollar figure the model displays. Without it, current-dollar GDP would be matched against a coefficient denominated in cheaper 2019 dollars, overstating jobs by the cumulative price growth since 2019.

6. Gambling-Specific Adjustments

Published IO tables contain only the blended sector 713 (Amusement, Gambling, and Recreation). The model applies adjustment factors derived from the BEA Detail IO tables (USEEIOv2.0.1-411) to isolate gambling-specific (NAICS 713200) coefficients:

Coefficient	Blended 713	Gambling (713200)	Ratio
Value added	0.651	0.577	0.89
Wages	0.406	0.255	0.63
TOPI	0.058	0.035	0.61

These ratios are applied to state-level blended multipliers to produce gambling-specific direct coefficients. Indirect and induced effects reflect supply-chain and household spending across other sectors, so no gambling adjustment is applied to those components.

7. Tax Revenue Estimation

The model computes three categories of tax revenue:

7.1 Taxes on Production and Imports (TOPI)

Derived from the IO tables, TOPI flows through the Leontief inverse like wages and captures sales, property, excise, and other production-related taxes across all direct, indirect, and induced activity.

7.2 Gaming Tax

A state-specific tax on Gross Gaming Revenue (GGR), sourced from state gaming commission reports and the AGA State of the States. Supported structures:

- **Flat rate** — a single percentage (e.g., Nevada at 6.75%)
- **Graduated tiers** — increasing rates at revenue thresholds (e.g., Colorado, Iowa)

- **Split by game type** — different rates for slots vs. table games (e.g., Pennsylvania)
- **Split-tiered** — separate graduated schedules for slots and tables (e.g., Illinois)

7.3 Payroll and Household Taxes

Employer-side payroll taxes (FICA, FUTA, SUTA, and state-specific SDI/PFML contributions) are applied to estimated wages. Household income taxes on supported wages are estimated via BEA personal current tax ratios (federal, state, and local).

8. Data Sources

Source	Vintage	Use
EPA State Input-Output (StateIO) models, via the stateior R package	2019	State multipliers (pre-pandemic year avoids distorted 2020 relationships)
BLS Quarterly Census of Employment and Wages (QCEW)	2023 annual averages	Employment and wage coefficients
BEA Detail IO tables via useeior (USEEIOv2.0.1-411)	2017	Gambling-specific adjustment factors
State gaming commissions, AGA State of the States	Current	Gaming tax schedules
BLS CPI-U annual averages	Current	Inflation adjustment

Multipliers vary significantly across states due to differences in economic structure, supply-chain density, and trade patterns. States with more diversified, self-sufficient economies tend to have higher multipliers because more supplier spending stays in-state.

9. Sector and Property-Type Mapping

Revenue entered by department is mapped to the corresponding IO sector with state-specific multipliers:

IO sector	NAICS	Casino department
711AS	711, 712	Other (entertainment, retail)
713	713	Gaming (GGR)
721	721	Lodging
722	722	Food & Beverage

When total revenue is entered instead, the model applies property-type-specific multipliers. Each property type carries its own multiplier set reflecting its economic structure:

NAICS	Property type	Typical operations

721120	Casino Hotel	Full-service resort with gaming, lodging, F&B;
713210	Stand-alone Casino	Gaming facility without hotel
713290	Slot Parlor	Slot machines / VLTs only
722410	Bar/Restaurant Gaming	Gaming secondary to food/beverage service
—	Online Casino / Sports Betting	Remote gaming operations (separate multiplier sets)

10. Limitations and Appropriate Use

- Results are **estimates** based on average inter-industry relationships; they should be interpreted as indicative rather than definitive.
- State-level analysis captures only impacts within the state boundary; spending that “leaks” to other states is excluded. This yields conservative estimates of local benefits.
- IO models assume fixed production technology and no supply constraints; very large projects may alter local economic structure in ways the model does not capture.
- The model is appropriate for policy analysis, planning, and comparative assessment. It is **not** appropriate for precise forecasting, investment decisions without professional advice, or legal proceedings.

11. Versioning

The model version is the release edition year (e.g., GEMS 2026). When a new edition is published — whether from regenerated datasets or methodology updates — the version is bumped and this document is updated. Reports generated by the calculator embed the model version and the data vintages used, so results remain traceable.

Version	Released	Notes
GEMS 2026	2026	Initial release. 2019 EPA StateIO multipliers, 2023 QCEW employment, USEEIOv2.0.1-411 gambling adjustments.

12. References

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About the Author

Dr. Kahlil Philander is an economist specializing in the analysis of policy and consumer behavior in the gaming industry, with nearly 20 years of applied research experience in economic impact measurement across academia, industry, and government. He is a tenured Associate Professor at Washington State University's Carson College of Business, holds a Ph.D. in Hospitality Administration from the University of Nevada, Las Vegas, and an M.A. in Economics from the University of Toronto. His research portfolio includes 40 peer-reviewed publications alongside dozens of industry reports.

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